

Mapping the Wizards' Path: A Systematic Review of Wizard-of-Oz in HCI

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Abstract

The Wizard-of-Oz (WoZ) method has long been a core prototyping technique in Human-Computer Interaction (HCI), in which users interact with systems that seem autonomous but are actually controlled by hidden human operators. Advances in interactive technologies have expanded the landscape of future system behaviors, broadening both where and how WoZ is used. However, as more envisioned behaviors become technically feasible, the distinction between engineering a system and simulating an interaction becomes blurred, making it essential to clarify when and why to employ wizarding. This paper presents the first systematic review of WoZ in HCI, drawing on 194 papers from SIGCHI venues to identify ten application domains, five wizard control types, eight motivations, and five categories of concerns. Building on these findings, we propose a reciprocal evolution framework that interprets how technology and wizarding shape each other, and derive guidelines for the rigorous application of WoZ. We further illustrate the framework through emerging prototyping practices with Large Language Models (LLMs).

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

Wizard-of-Oz, Systematic review, Prototyping

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1 Introduction

Wizard-of-Oz (WoZ) is a prototyping method in which users interact with a system that appears autonomous, while its key functions are covertly executed by human operators [18, 69]. A well-known early illustration is the "listening typewriter" [41]: participants believed that they were dictating to an automatic speech-recognition system, yet every word was actually typed in real time by a hidden typist. By simulating system behaviors that real technologies cannot yet support, WoZ enables researchers to approximate future interactions before they are technically feasible.

As interactive technologies have continued to advance, the horizon of "future systems" has expanded in parallel. Consequently, WoZ has been adopted across more domains to prototype a broader set of system operations. This trend is visible across areas ranging from autonomous driving [17, 128], smart home [81, 107] to virtual reality [1, 36], where "wizards" orchestrate multimodal sensing [19], context interpretation [76], physical actuation [34], and even blend their control with automated agents [26].

At the same time, some behaviors that previously required human wizarding have become increasingly feasible to support with current technologies. For example, early "listening typewriters" have been gradually replaced by Automatic Speech Recognition (ASR) [116]. Furthermore, Large Language Models (LLMs) can now handle content generation [158], reasoning [48], and planning [25] tasks that earlier WoZ prototypes enacted manually. In this context, automation complicates the decision on what to implement with available technologies and what to simulate through human control, requiring more deliberate consideration regarding prototyping method choices.

These trends show the need for a clearer map of how WoZ is employed across HCI. Despite the long-standing use of WoZ, previous reflections remain scattered across individual domains and case studies, offering little cumulative understanding [114, 119, 143]. Prior attempts to review WoZ include Riek's survey [122] in Human-Robot Interaction (HRI), which remains influential for its methodological rigor and reporting guidelines. However, the narrow scope in robotics and a decade-old corpus (2001–2011) limit its applicability for understanding WoZ's broader use in contemporary HCI. Therefore, a systematic literature review is needed to consolidate

the dispersed body of work, identify recurring patterns in WoZ practices, and reveal the motivations and methodological considerations that guide its use.

To this end, this paper presents the first systematic review of WoZ in HCI. Our review spans the broader HCI landscape, covering 194 full papers published across major SIGCHI venues from 1988 to 2025. Specifically, we focus on addressing the following Research Questions (RQs):

RQ1 (What): *What domains has WoZ been applied in HCI?*

RQ2 (How): *How do wizards control systems in WoZ studies?*

RQ3 (Why): *Why do researchers employ WoZ in HCI studies?*

RQ4 (Trade-offs): *What methodological concerns and mitigation strategies are reported in WoZ studies?*

Drawing on deductive and inductive coding of the corpus, our analysis reveals that WoZ has been applied in ten domains, including *Work & Creativity Support*, *Human-Robot Interaction (HRI)*, *Mobile & Ubiquitous & Wearable Computing*, *Automotive User Interfaces*, *Multimodal & Emerging Interaction Techniques*, *Mixed/Virtual/Augmented Reality (XR/VR/AR)*, *Education*, *Health & Well-being*, *Accessibility & Aging* and *HCI Methods & Tools*. We also identify five types of wizard control, spanning from *Communication Control* and *Embodied & Environmental Action Control* to *System Logic Control*, *Pre-scripted Content Control*, and *Wizard Collaboration & Mediation*. Through extracting authors' stated reasons for employing WoZ, we synthesize eight distinct motivations, including overcoming *Technical Limitations*, *Controlling Confounds*, supporting *Adaptive Flexibility*, creating *Psychological Realism*, achieving *Cost-Efficiency*, *Aligning with Prior Works*, *Benchmarking*, and following *Safety & Ethical Constraints*. Finally, we examine the methodological trade-offs reported in WoZ studies, detailing five categories of concerns: *Temporal and Performance Constraints*, *Wizard Variability and Bias*, *Ecological Validity*, *Deception and Ethics*, and *Transparency and Reproducibility*, alongside some mitigation strategies researchers employ to address them.

Building on these findings, we synthesize a broader methodological framing that characterizes how WoZ functions within today's prototyping landscape. This framework delineates the reciprocal evolution of technology and WoZ, where advances in interactive technologies and wizarding practices together expand the space of system behaviors that prototyping can explore. We then propose guidelines for employing WoZ, clarifying the kinds of research aims it can support and the practical principles that researchers should anticipate when using it. Finally, these insights are situated within the context of LLMs, detailing both different usage and emerging wizard-LLM co-use patterns that further demonstrate how WoZ remains relevant alongside increasingly capable technologies.

Here we list our contributions:

- We present the first large-scale systematic literature review of WoZ in HCI, based on a corpus of 194 full papers published across SIGCHI venues from 1988 to 2025. Our coded corpus has been released as an open dataset (<https://github.com/NICE-HKU/WoZ-lib>), supporting transparency and reuse.
- We provide a consolidated map of WoZ practices, identifying ten application domains, five wizard control types, eight motivation types, and five concern types.

- We synthesize a methodological framework describing the reciprocal evolution of technology and WoZ, offering practical guidelines for employing WoZ.
- We contextualize WoZ within the emergence of LLM-based prototyping, identifying key differences and intersections between wizarding and automated intelligence in contemporary prototyping practice.

2 Related Work

This section situates our work in relation to prior research. We first review the origins and diverse applications of WoZ in HCI, then summarize existing reflections on its strengths and limitations, and finally discuss how prototyping tools and automation have evolved to scale WoZ practices and support the simulation of increasingly intelligent systems.

2.1 Wizard-of-Oz: Origins and Applications

The methodological roots of WoZ in HCI can be traced back to the early 1980s. Gould et al. [41] famously demonstrated the utility of human simulation with their "listening typewriter". Kelley [69] subsequently formalized the technique as an iterative design methodology. Named after the Wizard of Oz metaphor, the method's defining characteristic is the simulation of system intelligence without disclosing the presence of a human intermediary [18], distinguishing it from broader human-in-the-loop [20] or teleoperation [53] approaches where human involvement is explicit.

Early applications of WoZ focused on natural language interfaces. For example, Qvarfordt et al. [117] used WoZ to manipulate the human-likeness of spoken feedback in a timetable system to examine how user experiences relate to their preferences. Over time, WoZ became a widely adopted HCI method as interaction technology advanced, extending from ubiquitous computing [81] to education [21], and more recently to domains such as HRI [175], automotive user interfaces [31], and Virtual Reality (VR) [1]. Riek's systematic review [122] of 50 WoZ studies in robotics between 2001 and 2011 demonstrates the methodological diversity and increasing prominence of WoZ in prototyping autonomous behaviors.

While prior work has traced WoZ's origins and diverse applications across domains, these accounts are scattered and lack a systematic synthesis within HCI. Our review fills this gap by consolidating 194 WoZ studies across SIGCHI venues, enabling a comprehensive mapping of where and how WoZ has been employed.

2.2 Evaluating WoZ: Strengths and Limitations

The appeal of WoZ lies in enabling rapid, high-fidelity prototyping before technologies are mature. By letting human operators perform core functions, researchers can study authentic interactions at low cost. In healthcare, clinicians have acted as chatbots to evaluate conversational agents [85]. In sensitive settings such as social robotics [39] or child-computer interaction [30], WoZ allows safe early-stage evaluation while preserving user experience.

At the same time, WoZ raises several well-documented limitations. Operator-mediated interactions are prone to bias and inconsistency, as different wizards may vary in judgment or responsiveness, undermining data reliability [135]. Participants may overestimate system autonomy, and debriefings can alter their perceptions [151].

Additionally, WoZ involves deception, raising concerns about informed consent and trust that persist even after participants are debriefed [132]. Finally, reproducibility is limited, as many WoZ studies lack sufficient detail on operator behavior, decision rules, or logging procedures [123]. As Riek [122] argues, the lack of standardized reporting exacerbates these concerns, prompting calls for greater transparency.

Existing reflections on WoZ's strengths and limitations remain fragmented and anecdotal. To address this gap, our review extracts authors' own statements on motivations and concerns of using WoZ, and meta-synthesizes them into higher-level themes, offering a more transparent account of methodological knowledge.

2.3 Scaling WoZ: Prototyping Tools and Automation

Early Wizard-of-Oz studies relied entirely on manual operation. This approach placed a heavy cognitive load on the human operator and often caused inconsistency in system responses. To address this issue, HCI researchers developed specialized tools to support wizards. For example, Suede [73] used a visual interface to help designers structure voice interactions. Topiary [89] allowed wizards to simulate location changes on a map for context-aware applications. These tools reduced the burden on human operators by handling system logic and visualizing complex contexts.

Researchers also explored blending human effort with automated algorithms to improve efficiency. Dow et al. [26] introduced a mixed-initiative approach for controlling embodied characters, where a system suggested potential responses and a human wizard selected the best one, improving response speed and consistency. Similarly, crowd-powered systems like Chorus [80] distributed the wizard role among multiple workers to handle complex tasks.

Recent advances in LLMs have further shifted this spectrum. An LLM is a transformer-based Artificial Intelligence (AI) system trained on massive text corpora to generate human-like language for tasks such as translation, summarization, and dialogue [158]. This breakthrough renewed HCI interest, with recent work adopting models like GPT-4 or Claude as core engines for interactive prototypes [113]. For example, PaperWeaver [82] used an LLM to infer researchers' interests from their personal paper collections and automatically recommend relevant literature. Tools like SpeakEasy [7] also use generative models to suggest content for wizards.

On one hand, automated prototypes offer unprecedented scalability and realism [130]. On the other hand, they lack the experimental control that human wizards provide. Risks of hallucinated or unstable outputs could undermine reliability [177]. This creates a methodological ambiguity regarding whether to use human simulation or automated implementation. We systematically map how WoZ is applied today and synthesize emerging patterns of collaboration between wizards and AI models, clarifying how WoZ and emerging automation can be combined or contrasted in prototyping practice.

3 Review Method

This section describes our review method, including how we built the corpus and how we coded the papers to answer our research questions.

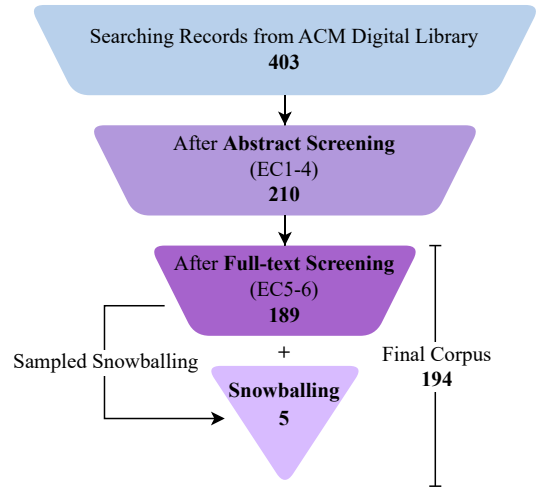


Figure 1: PRISMA-style flow diagram of corpus development. A total of 403 records were initially retrieved from the ACM Digital Library. After abstract screening (EC1–4), 210 papers remained; after full-text screening (EC5–6), 189 papers were retained. Sampled snowballing identified 5 additional papers, resulting in a final corpus of 194 WoZ studies.

3.1 Corpus Development

The corpus for this study was systematically developed through a process comprising a search, a two-stage screening, and a snowball sampling. Figure 1 presents the PRISMA flow chart for the entire development procedure.

We constructed the initial paper corpus by performing a rule-based search in the ACM Digital Library [147]. To comprehensively investigate the applications of the WoZ method across all subfields under the broad umbrella of HCI, we focused on all SIGCHI conferences with archival records [109], as SIGCHI represents the leading research community in this domain. The search terms were specified to cover different expressions and abbreviations of the method, with "wizard", "woz", "woo (WoO)", or "wooz (WoOz)" required to appear in the paper's title, abstract or keywords. The initial search returned 403 relevant records, which were subsequently screened in two stages: first based on abstracts, and then based on full texts.

During the abstract screening stage, each paper's title, abstract, and other basic information were examined, and entries that did not meet the criteria were removed according to the following Exclusion Criteria (EC):

- EC1: Papers that were not full publications or did not present a complete study (e.g. posters, demonstrations, workshops, or extended abstracts) were excluded.
- EC2: Papers whose titles or abstracts contained the keywords, but the context indicated that these terms did not refer to the Wizard-of-Oz method, were excluded.
- EC3: Papers whose abstracts indicated that the study merely used a dataset with "WOZ" in its name (e.g., the "DAIC-WOZ" corpus [42]) were excluded.

- EC4: Papers whose abstracts mentioned the WoZ method only in the background or in statements of future work were excluded.

For EC1, we restricted our review to full papers to ensure consistency in quality, as shorter formats typically lack the methodological or empirical detail necessary to address our **RQs**. EC2, EC3, and EC4 were applied to preliminarily exclude papers that did not actually implement the WoZ method. A total of 210 entries were discarded after the abstract screening, leaving 193 entries for the subsequent full-text screening.

The full-text screening was conducted in parallel with the coding process. Three researchers independently reviewed the full texts of all papers from the abstract screening, marking those deemed unsuitable for the final corpus. Subsequently, all contested entries were compiled and discussed, leading to the formulation of the following additional EC:

- EC5: Papers that proposed artifacts supporting the WoZ method, or presented literature reviews that mentioned WoZ method, but did not employ the WoZ method in an empirical study were excluded.
- EC6: Papers that used WoZ solely for corpus collection in Natural Language Processing (NLP) studies, rather than for HCI research, were excluded.

We decided to adopt EC5 because such papers do not provide empirical evidence of WoZ in practice, and therefore cannot contribute to answering our **RQs**. Likewise, papers that meet EC6 criteria fall outside the scope of our review due to their lack of engagement with HCI contexts. At this stage, our corpus was further refined to 189 papers.

At the same time, a sampling-based snowballing procedure was conducted. Snowballing refers to a literature search technique in which the reference lists of the included papers (i.e., backward snowballing), as well as papers citing them (i.e., forward snowballing), are examined to identify additional relevant studies [103]. From the 189 papers that passed the full-text screening, we sampled one quarter (47 papers) and conducted both backward and forward snowballing, restricted to the SIGCHI conference scope defined in the search and filtered by the exclusion criteria EC1–EC6. The result identified 5 additional relevant papers. This corresponds to a small corpus-level increment of 2.6% (5/189), suggesting high coverage.

Ultimately, our corpus comprised 194 HCI conference papers that reported empirical studies employing the WoZ method. This final set of papers served as the foundation for our subsequent systematic review and analysis. It should be noted that our final paper corpus does not cover every SIGCHI conference, since some conferences contained no papers that met our criteria after the search, two-stage screening and snowballing.

3.2 Coding Process

Based on the final corpus, we conducted a systematic coding process to examine how the papers relate to our research questions. Our approach combined both deductive and inductive strategies [4]. For **RQ1** and **RQ2**, we drew upon existing taxonomies from prior HCI literature as initial codebooks and iteratively refined them to suit the WoZ context. For **RQ3** and **RQ4**, we adopted an inductive approach, first extracting relevant excerpts from the papers and

then grouping them into emergent categories. Throughout the process, three researchers were involved in coding, discussion, and consensus building, ensuring both rigor and reliability.

In addressing **RQ1** (application domains) and **RQ2** (wizard control types), the initial codebooks were respectively seeded with Pang et al.'s LLM-ification review [113] and Riek et al.'s WoZ in HRI review [122]. We applied this codebook to randomly selected batches of papers (20 papers for each batch) in the corpus, and in each iteration considered two types of adjustments: (1) introducing new categories when papers could not be adequately classified, and (2) splitting overly broad categories into finer-grained subcategories and recoding previous papers. Each iteration involved three researchers independently applying the codebook, after which IRR (inter-rater reliability) was measured with Krippendorff's α [95]. Discrepancies were resolved through consensus discussions. After several rounds of refinement with consistent agreement among coders, the codebook was consolidated by merging redundant categories and eliminating those not represented.

With respect to **RQ3** (motivation of employing WoZ) and **RQ4** (concerns in WoZ studies), we employed an inductive, bottom-up coding procedure, given the lack of established frameworks for these topics. Relevant excerpts were systematically extracted from each paper, focusing on sentences that explicitly described the reasons for employing WoZ or mentioned the concerns of using WoZ. These excerpts were first open-coded independently by the three researchers, with initial codes generated directly from the textual evidence. In subsequent rounds, similar codes were iteratively grouped, refined, or elevated into higher-level categories, while divergent cases were examined to capture heterogeneity across studies. The IRR was also assessed and disagreements were resolved through consensus discussions.

The coding procedures resulted in a set of finalized codebooks covering domains, control types, motivations, and concerns. To ensure transparency and reproducibility, we preserved the original text excerpts used for coding and maintained a decision log documenting how categories were added, merged, or refined throughout the process. The average IRR scores for the final coding rounds were as follows: domains = 0.89, control types = 0.81, motivations = 0.82, and concerns = 0.90. Table 1 presents the finalized codebook definitions for our four research questions, providing the foundation for the analyses presented in the subsequent Results section.

4 Results

This section presents the key results of our systematic review. Figure 2 illustrates the temporal distribution of publications in our corpus, showing a steady increase of WoZ studies across SIGCHI venues. Building on this trend, we next detail our findings on application domains (**RQ1**), wizard control types (**RQ2**), motivations for employing WoZ (**RQ3**), and concerns & mitigation of WoZ (**RQ4**).

4.1 Application Domains

Across the 194 papers, we identified ten distinct domains where WoZ has been applied. In the following subsections, we elaborate on each domain with representative examples and highlight how WoZ has been leveraged to support their research goals.

	Code	Definition	Examples
Application Domains	<i>What domains has WoZ been applied in HCI?</i>		
	Work & Creativity Support	Studies on systems that support productivity, collaboration, and creative practices.	[1, 15, 28]
	HRI	Studies on social, collaborative, or embodied interaction with robots.	[22, 50, 104]
	Multimodal & Emerging Interaction Techniques	Studies on novel input/output modalities beyond standard interfaces.	[14, 36, 164]
	Mobile & Ubiquitous & Wearable Computing	Studies on interactive systems embedded in mobile, pervasive, or wearable contexts.	[3, 14, 131]
	Automotive User Interfaces	Studies on in-vehicle interactions and autonomous driving scenarios.	[66, 92, 157]
	XR/VR/AR	Studies on mixed, virtual, or augmented reality environments.	[31, 36, 178]
	Education	Studies on technologies that support teaching and learning.	[6, 30, 54]
	Health & Well-being	Studies on clinical workflows, health tools, and well-being interventions.	[65, 108, 146]
	HCI Methods & Tools	Studies using WoZ to advance prototyping and evaluation practices.	[89, 90, 134]
	Accessibility & Aging	Studies on supporting people with disabilities and aging populations.	[153, 174, 183]
Wizard Control Types	<i>How do wizards control systems in WoZ studies?</i>		
	Communication Control	Wizard manages verbal and non-verbal interaction between the user and the system.	[57, 86, 138]
	System Logic Control	Wizard simulates the internal perception–decision processes of an intelligent system.	[30, 148, 163]
	Embodied & Environmental Action Control	Wizard manipulates agents' bodies, external devices, or environmental cues.	[34, 92, 171]
	Pre-scripted Content Control	Wizard triggers predefined responses or assets.	[62, 88, 159]
	Wizard Collaboration & Mediation	Wizard coordinates with other wizards or automated components.	[76, 84, 154]
Motivations of Employing WoZ	<i>Why do researchers employ WoZ in HCI studies?</i>		
	Technical Limitation	WoZ substitutes for system capabilities that are technically infeasible or not yet developed.	[67, 79, 173]
	Alignment with Prior Work	WoZ is employed in line with established research practices.	[52, 77, 121]
	Adaptive Flexibility	WoZ offers adaptive system behaviors and grants users greater freedom of interaction.	[124, 154, 178]
	Psychological Realism	WoZ makes users believe they interact with autonomy.	[43, 142, 173]
	Controlling Confounds	WoZ eliminates or manipulates potential confounds to preserve study validity.	[28, 63, 176]
	Cost-Efficiency	WoZ avoids resource-intensive system development through rapid prototyping.	[39, 111, 159]
	Ethical & Safety Constraints	WoZ in domains where testing autonomous systems poses significant safety or ethical risks.	[23, 31, 128]
Concerns & Mitigation of WoZ	Benchmark	WoZ provides a benchmark for evaluating other systems or design alternatives.	[85, 126, 131]
	<i>What methodological concerns and mitigation strategies are reported in WoZ studies?</i>		
	Temporal and Performance Constraints	WoZ performance is limited by human latency or errors that disrupt interaction flow.	[15, 16, 120]
	Wizard Variability and Bias	WoZ consistency is compromised by variations in different wizards.	[46, 112, 153]
	Ecological Validity	WoZ simulations cannot fully reproduce the autonomous systems.	[2, 131, 182]
	Deception and Ethics	WoZ deception raises ethical concerns regarding informed consent and participant trust.	[19, 52, 61]
	Transparency and Reproducibility	WoZ reproducibility is hindered by insufficient reporting of wizard protocols and logs.	[78, 122]

Table 1: Finalized codebook definitions for four Research Questions (RQ1–RQ4)

4.1.1 Work & Creativity Support (27.8%, N=54). This domain encompasses WoZ studies investigating how interactive systems can augment human productivity, collaboration, and creative practices. WoZ has long supported this domain, ranging from early creativity workshops [156] and digital filmmaking support tools [19] to more recent studies of embodied design collaboration [74], VR-based co-creation [118], and agent-led teamwork scenarios [138].

For example, GestureWiz [141] combined WoZ and crowdsourcing to build a human-powered gesture design environment, enabling designers to iteratively prototype and refine gesture vocabularies in real time. Similarly, a long-term WoZ deployment study introduced the Co-Performing Agent [70], where wizards controlled an assistive agent to explore how collaborative performance systems could sustain user engagement over two months. More recently,

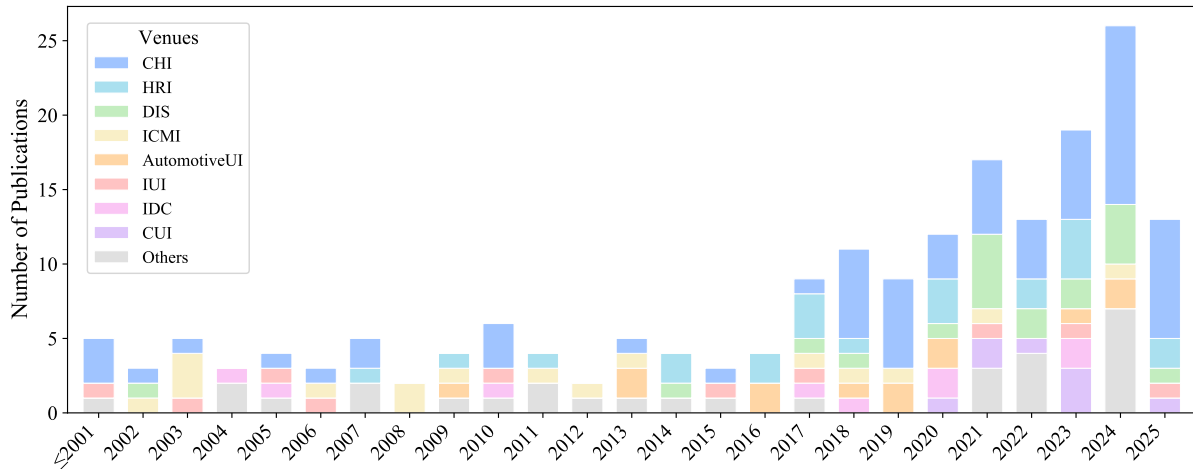


Figure 2: Distribution of WoZ publications across years and venues. Venues are grouped by total publication frequency into the top eight series (CHI, HRI, DIS, ICMI, AutomotiveUI, IUI, IDC, and CUI) and “Others,” with counts for 2025 including publications up to June only.

the SpeakEasy system [7] employed a hybrid WoZ–LLM workflow, with wizards producing speech content while generative models recommended appropriate speaking styles.

4.1.2 Human-Robot Interaction (HRI) (18.0%, $N=35$). This domain covers WoZ studies that examine how people interact with robotic systems in social, collaborative, and embodied contexts. Studies in this domain span a broad spectrum: from investigations of adaptive affective displays in child–robot encounters [149] and early explorations of voice-controlled drones [10], to more recent work on parent–child–robot collaborations around learning activities [54], robot-assisted craft making [148], and even experiments with scent as a novel channel for social interaction [104]. For example, one study in a museum tour context [35] used WoZ to control a robot’s opening interactions with visitors, identifying behavioral patterns and deriving multiple adaptable strategies. Another work simulated a service robot handling malicious complaints [101], demonstrating how scripted responses shaped perceptions of diligence and informed behavioral models. Moving to an elevator in an office building, a WoZ experiment [34] compared human-like and machine-like strategies of a social robot, assessing their effects on perception, comprehension, and efficiency.

4.1.3 Multimodal & Emerging Interaction Techniques (13.9%, $N=27$). This domain highlights WoZ studies that explore novel interaction modalities and experimental technologies beyond traditional interfaces. Early work mainly examined speech-based multimodality, such as simulating alternative voice feedback styles [117], and model the integration of speech and pen input under different error conditions [110]. Subsequent studies extended to gesture and physical modalities. One study simulated motion-gesture recognition under controlled error patterns [68]. Other efforts examined how everyday objects could be appropriated as programming inputs [59]. Recent work has further investigated wearable and sensory modalities, which sometimes overlap with other domains.

Examples include the SuspenderMender [108] prototype that simulated supportive touch to alleviate student anxiety, the use of gait gestures as input for mobile AR interfaces [168], and olfactory cues to emulate robotic smell detection [104].

4.1.4 Mobile & Ubiquitous & Wearable Computing (12.9%, $N=25$). This domain captures WoZ studies investigating interactive systems embedded in everyday mobile, pervasive, or wearable contexts. Researchers have used WoZ to prototype location-enhanced services [89], smartwatch feedback modalities [40], and wearable input techniques such as micro-gestures [14] and gait gestures for AR interfaces [155]. For example, one study applied WoZ in the living room to simulate natural gesture control of digital media, enabling analysis of how users coordinated remote gestures with system feedback [81]. Another investigation used a WoZ-controlled deformable dining-table decoration to study how material transformations shape interaction during shared meals [102]. In a different setting, researchers built a multimodal kitchen assistant prototype where wizards provided proactive voice and visual guidance, allowing systematic examination of user responses in complex cooking tasks [181].

4.1.5 Automotive User Interfaces (12.4%, $N=24$). In the automotive domain, WoZ has been widely employed to study interactions with in-vehicle systems and autonomous driving scenarios. One line of work focuses on dialog and information presentation, such as simulating natural language versus command-based speech interfaces in driving simulators [46], testing broadcast-style information delivery [93], and introducing conversational explanations in autonomous driving error scenarios [136]. Another stream examines automated driving behaviors and user responses, where hidden drivers controlled vehicles to compare different driving styles [173], to study pedestrian reactions in different cities [17], and to manipulate stopping distances at crosswalks [100]. A third direction develops prototyping infrastructures for automotive research, such as the

“WoZ Way” system for remote in-car control [94], theater-system techniques for highly automated scenarios [133], and studies of sound interaction design in public transport contexts [115].

4.1.6 Mixed/Virtual/Augmented Reality (XR/VR/AR) (11.9%, N=23). This domain investigates WoZ applications in extended reality environments, including mixed, virtual, and augmented reality (i.e., XR, VR, and AR). Notably, these studies often intersect with other domains, such as *Work & Creativity Support* and *Education*. With the former domain, WoZ supports immersive content creation and collaboration, including the use of paper-based VR prototyping in 360proto [106], cross-AR/VR orchestration in XRDirector [105], and eliciting user expectations for conversational programming agents in VR [1]. With the latter, WoZ enables social and contextual applications, such as deploying the Big Buddy moderator in a social VR classroom [30] and providing asynchronous learning support through AR in collaborative drawing tasks [31].

4.1.7 Education (11.3%, N=22). This domain examines WoZ studies investigating how educational technologies can support teaching and learning processes. Applications can be broadly grouped into two strands in terms of learning content. The first strand focuses on supporting skill and knowledge learning. Early work compared polite versus direct strategies of a WoZ-driven tutoring agent in computer-based instruction [161]. Later studies applied WoZ to subject-specific learning, such as tutoring children’s mathematics through a dialog-based chatbot [127] or enabling math-related parent–child interactions at home with a voice-based game [171]. The second strand emphasizes language and storytelling. Examples include a WoZ-controlled American Sign Language learning game for deaf children [51], a simulated child–robot system for collaborative story narration [145], and StoryCoder [24], which prototyped a speech-based environment for children’s creative writing and computational thinking.

4.1.8 Health & Well-being (8.8%, N=17). This domain addresses WoZ studies that simulate health-related systems to examine user needs, clinical practices, and well-being interventions. The use cases can be distinguished by their settings, ranging from hospitals to everyday health support. In clinical environments, examples include a WoZ-based display in emergency care to examine patient information needs [166], a multimodal conversational agent used in mental health interviews to capture verbal and non-verbal signals [146], and pre-consultation chatbots benchmarked against LLM systems for adaptive questioning [85]. In daily scenarios, studies range from mobile tools for activity tracking [111] and personalized coaching for exercise routines [157] to probes identifying psychological states behind problematic smartphone use [169]. Other projects explored well-being through social and emotional support, such as robot companions in pediatric wards [65], adaptive lighting to sustain social connection [55], and wearables designed for shared anxiety management [108].

4.1.9 HCI Methods & Tools (8.8%, N=17). This domain consists of studies that use WoZ not as an end application but as a methodological instrument for advancing prototyping and evaluation practices. Early work introduced Topiary [89], a prototyping tool that relied on WoZ techniques to simulate location-enhanced applications.

Building on such tools, subsequent studies examined how new input techniques could be assessed through WoZ setups. For example, DirectionalCrossing and Steering were experimentally analyzed to reduce interaction costs in continuous input tasks [90]. Researchers have also reflected on the boundaries of the method itself: the so-called Oz of Wizard approach [143] highlighted how WoZ can simulate human participation in scenarios where direct deployment of autonomous systems is infeasible or unsafe. Additionally, efforts such as WebWOZ [134] consolidated these insights into a generalizable framework, combining scenario prototyping, wizard observation, and component integration to support language technology research.

4.1.10 Accessibility & Aging (7.2%, N=14). This domain covers WoZ studies designed to investigate how interactive systems can better support people with disabilities and aging populations. For deaf and hard-of-hearing users, studies simulated ASL-based educational games [51] and personal assistants capable of sign language input [37, 153]. For blind and low-vision users, researchers explored interactive 3D printed models [121], chart Question and Answer (Q&A) systems for data visualization [71], and smartwatch prototypes offering auditory or tactile feedback to improve everyday awareness [40]. In parallel, voice-based assistants were tested to compare how older and younger adults query complex mobile interfaces [172]. Similarly, the Storybell system connected grandparents and children through remote narration [127].

4.2 Wizard Control Types

This section delves into the various control types wizards utilize in WoZ studies within the context of HCI. Starting from the specific control operations, we have abstracted five higher-level wizard control types: *Communication Control*, *Embodied & Environmental Action Control*, *System Logic Control*, *Pre-scripted Content Control* and *Wizard Collaboration & Mediation*.

4.2.1 Communication Control (55.2%, N=107). This type of control refers to the wizard’s management of the *verbal* and *non-verbal* elements of interaction between the user and the system. It includes generating written or spoken feedback (e.g., through text or speech synthesis) and using non-verbal cues (e.g., head movements or gaze) to guide or inform the user.

In Graphical User Interface (GUI) systems, *text* is a widely used modality for communication control. Many WoZ systems in research are implemented as traditional chatbots, where the wizard behind the scenes conveys information to users through pre-configured or dynamically edited text [86]. In contrast, direct *speech*-based communication is often leveraged to enhance system engagement and realism, typically through technologies such as Text-to-Speech (TTS) [60, 72, 138]. For example, Hubbard et al. [60] guided storytelling by having a wizard-simulated system engage in a Q&A dialogue with children. Some studies combine both modalities. Kuang et al. [74] examined UX evaluators’ preferences for speech versus text by having the Wizard impersonate an AI assistant, responding to participants’ questions in both formats during usability test video analysis.

Non-verbal communication control typically occurs when the wizard needs to simulate embodied actions, particularly when the

system must appear to understand and respond to the user in a more natural, human-like manner. Unlike *Embodied & Environmental Action Control*, which primarily focuses on controlling the agent's physical behavior, the primary aim of non-verbal communication control here is to convey information, rather than simply directing the agent's actions. For instance, Zhang et al. [176] used wizard-controlled multimodal signals, such as the agent's head movements and visual highlights, to simulate the agent's ability to provide real-time understanding and feedback. Similarly, Jaber et al. [62] used the WoZ method to simulate an intelligent agent capable of maintaining eye contact, revealing that gaze interactions can reduce task interruptions in human-robot collaboration.

4.2.2 System Logic Control (48.5%, N=94). This type of control captures cases where the wizard simulates an intelligent system's internal perception-decision processes. At this level, the wizard is responsible for interpreting user inputs, applying reasoning rules, and selecting appropriate system behaviors, thereby creating the impression of autonomy and adaptivity.

System logic control often spans both *perception* and *decision-making*, though individual studies may emphasize one aspect over the other. For example, Chen et al. [14] illustrated the perceptual role by having the wizard recognize participants' mouth micro-gestures and trigger corresponding smartphone operations. By contrast, E et al. [28] highlighted the decision-making role in a WoZ-driven visual design tool, where the wizard decided whether feedback should be delivered immediately in-action or in a delayed on-action manner. Integrating both dimensions, Oh et al. [107] simulated a smart home assistant in which participants, acting as wizards, inferred users' intentions and proactively provided services.

A special line of work applies system logic control for *error injection*, where the wizard deliberately introduces misrecognitions, delays, or inconsistencies. Such an operation is sometimes used to create controlled failure patterns; Katsuragawa et al. [68] employed WoZ to simulate a smartphone motion gesture recognition system, comparing the effects of different rejection strategies on user experience under the same overall rejection rate. In another example, Zhou et al. [182] introduced random error injection and temporary loss of control when the wizard operated supernumerary multi-arms in a VR environment, simulating potential system failures.

4.2.3 Embodied & Environmental Action Control (47.9%, N=93). This control type involves the wizard's direct manipulation of either an agent's body (*embodied action control*) or external devices and environmental cues (*environmental action control*) to simulate system actions.

Embodied action control generally involves manipulating, moving, or navigating a *robot* or *virtual character*. Tian et al. [148] employed WoZ to control a manipulator robot, enabling it to collaborate with participants in a cabin-building task by managing object handovers. In another study, Fink et al. [167] used a wizard-controlled "Ranger" robot toy box as a playful intervention to motivate children to tidy up their toys. Recent studies on virtual character control increasingly integrate VR to provide a more immersive user experience. For instance, Big Buddy [30], an embodied AI moderator introduced via the WoZ method in a simulated social VR

environment, was designed to explore how future child-friendly moderation systems could be implemented in virtual spaces. In addition, Zhang et al. [176] demonstrated a WoZ-controlled AI agent that combined non-verbal feedback with embodied actions, moving objects in a VR scene according to users' instructions.

Environmental action control encompasses the wizard's manipulation of external devices and environmental cues (e.g., light, sound, or scent feedback) to shape the user's interaction context. In Light Bridge [55], the wizard dynamically adjusted lighting intensity to simulate the location and activities of a remote user. Jain et al. [64] employed WoZ in a situated performance, where the wizard coordinated with actors' movements to produce corresponding auditory feedback in the room. Similarly, Nakashima et al. [104] enabled a wizard to control the release of scents backstage, simulating a robot's ability to detect and transmit olfactory information. Another special case of environmental action control appears in the context of *Autonomous Vehicles (AVs)*, where a wizard (also called a "ghost driver") is in fact responsible for operating the vehicle, while passengers or other traffic participants perceive it as being autonomously driven. In the study by Meurer et al. [97], three researchers acted as drivers of a vehicle disguised as an autonomous Robo-taxi, transporting passengers according to prearranged routes. Similarly, Moore et al. [99] conducted a field experiment in which a wizard drove a simulated driverless vehicle, providing an analysis of grieving behaviors directed at AVs by both pedestrians and other drivers.

4.2.4 Pre-scripted Content Control (36.6%, N=71). Pre-scripted content control refers to the wizard triggering pre-prepared responses or assets instead of generating real-time outputs. It manifests in two main ways. First, the wizard delivers system behavior by selecting from a set of prepared responses or assets, such as scripted dialogues or fixed interaction flows. This type of pre-scripted content control is frequently intertwined with other forms of control, such as *Communication Control* or *Embodied & Environmental Action Control*. For example, one study [138] had the wizard simulate an AI agent by selecting from multiple candidate responses to guide a team in completing a task. In another context, a cooking instruction study [181] relied on the wizard to manage the interaction by playing instructional videos and sharing the screen with participants.

In the second, the prepared content is created by human experts, with the wizard deploying it during the study to simulate a system that appears knowledgeable or authoritative. For instance, Hassan et al. [49] presented a dance movement recognition tool whose outputs were actually manually annotated prior to the empirical study. Similarly, CreepyCoCreator [118] highlighted that the content used for co-creation with users in the VR environment was prepared in advance by human artists.

4.2.5 Wizard Collaboration & Mediation (16.0%, N=31). This control type refers to settings where control is distributed across multiple wizards or between a wizard and an automated component, requiring coordination to maintain coherent interactions.

Multi-wizard collaboration arises when different wizards are assigned complementary roles that must be coordinated to sustain a seamless interaction. In the study by Zhang et al. [179], two wizards were employed: one was responsible for teleoperating the robot's movements, while the other controlled the robot's verbal

responses. In Kim et al.'s [72] study on AVs, one wizard was responsible for driving the vehicle, while another provided voice feedback in response to the guiding commands of the passengers.

On the other hand, an emerging direction involves hybrid WoZ settings in which wizards collaborate with LLMs, integrating human oversight with automated generation. Cheng et al. [15] demonstrated this collaboration by assigning the wizard to structure prompts and orchestrate interactions, while GPT-4 and DALL-E were leveraged to produce feedback and images. Another study presented SpeakEasy [7], where GPT-4o was employed for context comprehension and tone recommendations, while the wizard managed the delivery of outputs in different tones.

4.3 Motivations of Employing WoZ

We extracted and coded explicit statements of motivation from our corpus, synthesizing them into eight categories that capture the diverse rationales for employing WoZ. It should be noted that only 87.3% ($N = 150$) of the analyzed papers explicitly state a motivation.

4.3.1 Technical Limitation (45.9%, $N=89$). This motivation refers to using WoZ as a substitute for system capabilities that are technically infeasible or underdeveloped at the time of study. Among all identified motivations, this is the most prevalent, reflecting WoZ's role in overcoming algorithmic immaturity across diverse domains. For example, Zhang et al. [179] employed WoZ to compensate for the limitations of contemporary algorithms in achieving human-level understanding of Indirect Speech Acts (ISAs). Similarly, Kuang et al. [74] pointed out that even state-of-the-art AI techniques still struggle to reliably detect usability problems, making WoZ a practical substitute for studying evaluators' behavior. In a broader sense, Li et al. emphasized that their study was "not hindered by current technical limitations" [84] by leveraging WoZ.

4.3.2 Alignment with Prior Work (16.5%, $N=32$). This motivation refers to employing WoZ in line with established research practices. Studies often appeal to prior work when adopting WoZ, framing it as a well-established method that simultaneously aligns with other motivations such as *Technical Limitation* [74] or *Adaptive Flexibility* [154]. Typically, such papers include phrasing like "researchers have..." [100] or "...in prior research/studies" [13], explicitly situating their study within this tradition.

4.3.3 Adaptive Flexibility (14.4%, $N=28$). This motivation refers to the use of WoZ to allow researchers to adapt system behaviors flexibly and to provide users with greater freedom of interaction. By employing a human wizard dynamically controlling system responses, researchers can support naturalistic interaction and explore a wider range of design possibilities. For instance, Palani et al. [112] highlighted that WoZ enabled them to accommodate individual differences in how people approached complex information tasks. Reinders et al. "allowed participants to interact with 3D models in any way they felt natural" [121], while dynamically adjusting the model's agency through wizard control. Similarly, Weijdom [165] emphasized that WoZ affords flexibility in user testing by granting participants more freedom of expression than would be possible with automated systems.

4.3.4 Psychological Realism (12.9%, $N=25$). This motivation refers to using WoZ to create psychologically realistic interactions, where participants perceive the system as autonomous rather than a human interlocutor. This rationale explains why researchers disguise a human as a system instead of letting participants interact with a person, imagine scenarios, or use other indirect methods. For example, Glasser [37] pointed out that actual interaction with a sign-language assistant elicited behaviors that would not emerge in purely hypothetical settings. E et al. [28] demonstrated that when users assumed feedback came from a computational agent, they engaged more openly, free from typical social pressures. Similarly, Song et al. [140] found that participants unaware of the wizard's presence rated the robot more favorably and reported higher enjoyment.

4.3.5 Controlling Confounds (12.4%, $N=24$). This motivation refers to using WoZ to eliminate or manipulate potential confounding factors that could otherwise bias study results. Here, the focus lies in maintaining experimental validity, with a human wizard controlling system behavior to ensure that observed effects can be attributed to the intended design variables. For example, Hackenberg et al. [46] simulated two in-vehicle dialog systems using WoZ to maintain a constant recognition rate, isolating the impact of dialog design choices. Similarly, Jaber et al. [62] used a wizard to circumvent potential errors in speech recognition and gaze detection, ensuring that differences in user experience arose solely from the manipulated conditions. Zhang et al. [176] further leveraged WoZ to control when and how often misinterpretations occurred precisely, enabling systematic investigation of their influence.

4.3.6 Cost-Efficiency (12.4%, $N=24$). This motivation refers to the use of WoZ as a cost-effective strategy for prototyping and evaluation, avoiding the resource-intensive effort of developing fully functional systems. For instance, Viswanathan et al. [159] noted that building a dialogue system would require unavailable training data and extensive annotation, far beyond the scope of the study. Bouquet et al. [6] used WoZ to assess interaction concepts with AR and TV content quickly before committing to deeper development. More generally, Tsai et al. employed WoZ for gesture recognition just "to simplify development" [155].

4.3.7 Safety & Ethical Constraints (5.2%, $N=10$). This motivation refers to employing WoZ in domains where direct experimentation with autonomous systems could raise safety or ethical risks. It appears only in specific domains, such as AutomotiveUI and certain HRI studies. For example, Detjen et al. [23] noted that real-world testing with autonomous vehicles could endanger drivers, passengers, and other road users. Zhang et al. [179] used WoZ to study human-robot interaction while avoiding risks associated with autonomous motion control. In another study, Vtyurina and Fourney [160] adopted WoZ in contexts where privacy concerns made direct system deployment impractical.

4.3.8 Benchmark (3.6%, $N=7$). This motivation refers to using WoZ as a benchmark against which other systems or design alternatives can be evaluated. By positioning the wizard as a proxy for ideal or human-level performance, researchers can assess the gap between current technologies and aspirational capabilities. For example, Li et al. [86] used WoZ to represent the highest level of adaptability,

providing an upper bound for comparison. Similarly, Rouanet et al. [126] employed WoZ to approximate human-level vision and interpretation, thereby revealing how far robotic interaction could progress.

4.4 Concerns & Mitigation of WoZ

We also extracted and inductively coded authors' reported concerns with WoZ and the mitigation strategies they described. Across the 194 papers in our corpus, 30.4% ($N = 59$) mentioned such concerns, which we synthesized into five main categories.

4.4.1 Temporal and Performance Constraints (13.4%, $N=26$). A recurring concern is that wizard responses are slower or less consistent than those of automated systems, leading to noticeable latency, occasional typos, or disruptions in interaction flow. Such delays were frequently highlighted in conversational and embodied settings, where participants expected real-time responses [12, 74]. To mitigate these issues, researchers employed strategies such as operator training and rehearsal [22], pre-scripting common responses [28], or distributing tasks across multiple wizards [137]. Some studies introduced controlled delays, turning latency into a design variable [182].

4.4.2 Wizard Variability and Bias (8.2%, $N=16$). Another common concern is that different wizards may vary in judgment, typing style, or domain knowledge, introducing inconsistencies that undermine data reliability. In some cases, personal experience or interpretation was noted to bias how system awareness or user input was represented [153]. Mitigation strategies included standardized coding schemes [15] and constraining wizard actions to predefined options [28]. Several studies also adopted dual-wizard protocols or extensive training sessions to improve consistency across operators [22].

4.4.3 Ecological Validity (7.2%, $N=14$). Several studies noted that WoZ setups cannot fully reproduce the autonomous systems, raising concerns about how findings generalize to real-world deployments [104, 131]. The absence of authentic system errors, resource constraints, or failure modes may lead participants to overestimate system reliability or trust. To address this gap, researchers deliberately injected errors or delays, restricted wizard perception, or simulated partial system failures [182], approximating imperfect autonomy conditions.

4.4.4 Deception and Ethics (2.6%, $N=5$). Because WoZ relies on concealing the human operator, several papers raised concerns about informed consent and potential effects on participant trust [52, 61]. Some studies further noted the risk of participants misattributing human-like qualities to the system [78]. To mitigate these issues, researchers employed cover stories that framed the prototype as an early-stage technology [31, 52] and conducted thorough post-study debriefings [61]. A few works explicitly argued that limited deception is acceptable when necessary for eliciting naturalistic user behavior, provided participants are fully informed afterwards [61].

4.4.5 Transparency and Reproducibility (1.0%, $N=2$). Only a few papers explicitly reflected on the difficulty of replicating WoZ studies, noting that wizard actions, decision rules, and logging procedures were often underreported [78, 122]. This lack of detail complicates

comparisons across studies and limits methodological transparency. To mitigate such risks, some researchers adopted constrained protocols for wizard behavior and emphasized the importance of documenting operator actions in detail for future replication [34].

5 Discussion

Building on our review findings, this discussion synthesizes what WoZ contributes as a prototyping method in contemporary HCI. We first introduce a framework that explains how WoZ and technology co-evolve, then derive practical guidelines for when and how to employ wizarding. Finally, we situate these insights in the context of LLM-based prototyping, and conclude with limitations and future work.

5.1 The Reciprocal Evolution of Technology and WoZ

Across our corpus, we observe that technological progress does not make the WoZ method obsolete. Researchers continue to use wizards even after specific capabilities become automatable, shifting to simulate more advanced behaviors. Existing theories emphasize phasing the wizard out as systems mature [69], but they do not capture how technological advances also open new spaces where wizarding is reintroduced.

To explain this, we propose a framework that views WoZ and technology as forces that shape system behaviors together. This reciprocal evolution framework unfolds through three mechanisms that expand the plausibility horizon and the plausible space of interactive systems.

5.1.1 Technology Anchors the Plausibility Horizon. Foundational technologies define what systems can actually do in practice, and also shape what people expect from ordinary system behavior. These shared expectations define an outer boundary, which we refer to as *plausibility horizon*. This horizon represents the limit where users can maintain a suspension of disbelief [5]. Even when WoZ allows simulating capabilities that exceed the current state of the art, the credibility of such simulation is inherently constrained by this horizon. If the simulated behaviors exceed reasonable expectations too drastically, users may perceive the system as magic or deception [150]. Such perceptions can hinder the formation of coherent mental models and consequently compromise the ecological validity of the study [26].

Anchored by technology, the plausibility horizon is dynamic. As core sensing, inference, and actuation capabilities advance, the plausibility horizon also shifts outward, enlarging the set of behaviors that are credible for a system to exhibit. This expansion contextualizes our findings in Section 4.1 and 4.2. Earlier WoZ studies mainly simulated text [41] or speech [73] because these were the credible frontiers of intelligence at the time. However, as sensing and actuation became widespread, the plausibility horizon expanded to include embodied [26] and spatial [55] behaviors. In this sense, technological progress raises the baseline of what systems can reliably do, and consequently shifts wizarding toward simulating more complex behaviors that still lie within the plausibility horizon.

5.1.2 WoZ Explores the Plausible Space. The *plausible interaction space* (or *plausible space*) lies between the current engineering capabilities and the plausibility horizon. Technology grounds its inner boundary of what can already be engineered, while the plausibility horizon sets its outer boundary of what users still find credible. WoZ allows researchers to explore this space, enabling experience prototyping of future behaviors that are technically infeasible but conceptually coherent [8].

In practice, prototypes rarely rely on WoZ alone. Automated components handle the parts of the interaction that fall inside the inner boundary, while wizards build directly upon them, enacting behaviors that extend into the plausible space. Within this space, WoZ functions as a strategic filter for desirability [91]. It helps distinguish which imagined system behaviors meaningfully improve interaction, rather than merely extend what a system might theoretically do.

This role aligns with our findings in Section 4.3. Researchers employ WoZ to overcome *Technical Limitations* not only to enable basic system functionality, but also to support *Adaptive Flexibility* and create *Psychological Realism*. In addition, WoZ provides a human-performance upper bound *Benchmark* that future systems may aim to approximate. The interaction data collected during WoZ studies can serve as a reference for training future algorithms [42, 135]. By enacting interactions within the plausible space, WoZ allows designers to assess user expectations and interaction patterns long before engineering investment is committed. This approach helps determine whether we are building the right things, which concerns desirability, before we figure out how to build things right, which concerns feasibility [9].

5.1.3 The Cycle of Reciprocal Evolution. These mechanisms form a recursive cycle of *reciprocal evolution*. The advance of technological reality pushes the plausibility horizon outward, opening up a new plausible space. WoZ then migrates from the old space to this new space. Researchers move from simulating what once counted as future functionality to exploring the next frontier of interaction. As these new interactions are eventually automated, the plausibility horizon shifts again, and the next iteration of the cycle begins. This cycle can be understood as a process through which technology and WoZ jointly shape the evolving space of possible system behaviors. Positioning WoZ as a futuring method also foregrounds questions raised in recent analyses of HCI futuring: how uncertainty is explored, which time horizons are assumed, whose experiences are centered, and which dominant narratives are reproduced or contested [129].

The reciprocal evolution cycle helps explain the longevity of the method. As Kelley [69] articulated initially, and later emphasized by Riek [122], a valid WoZ process is iterative, where the human wizard is "gradually phased out" as autonomous features are "phased in". However, we argue that this obsolescence is specific to the capability being engineered. Once a given behavior is reliably automated, wizarding for that behavior can be reduced, but researchers often redeploy WoZ to probe new behaviors in the next plausible space.

However, we also acknowledge that this cycle operates within a theoretical boundary. It assumes that human cognitive or social capabilities still outpace automated systems in the domain of interest. As technologies advance toward Artificial General Intelligence

(AGI) [38] or super-human performance, this gap may narrow, or even invert. In this context, the challenge may shift from simulating the machine to elevating the human, considering how humans can harness such intelligence while maintaining agency and ethical control. The plausible space may then extend from imagining new system behaviors to imagining new forms of human augmentation. Thus, while our framework explains the historical and current trajectory of WoZ, how this reciprocal evolution adapts to the era of AGI remains an open question for future inquiry.

5.2 Practical Guidelines for Employing WoZ in HCI Research

Translating our findings into actionable implications, this subsection offers methodological guidance for the effective use of WoZ. We outline when WoZ best serves specific research goals and how to execute it rigorously once chosen.

5.2.1 Matching WoZ to Research Goals. The practical decision to employ WoZ requires aligning the method with specific epistemic goals. Our review of reported motivations (Section 4.3) reveals that researchers adopted WoZ for a wide range of purposes, indicating the need for a more systematic articulation of the conditions under which the method is well-suited. To this end, we synthesize three primary guidelines for identifying when WoZ constitutes the most appropriate methodological choice.

Guideline 1: prioritize WoZ for feasibility and safety checks.

When the target system capabilities remain technically infeasible or when deploying autonomous systems poses physical or psychological risks, WoZ serves as an essential methodological bridge. Our review indicates that overcoming *Technical Limitations* remains the primary motivation for employing WoZ, particularly for capabilities where algorithms still struggle with human-level nuance, such as interpreting indirect speech acts [179] or detecting subtle usability problems [75]. This practice aligns with Houde and Hill's foundational model of prototyping [56], which advocates for decoupling a system's "role" (i.e., its function and value to the user) and "look and feel" from its "implementation" (i.e., how it is built).

Furthermore, in high-stakes scenarios such as autonomous driving, WoZ acts as a necessary safety valve, allowing researchers to investigate user behaviors without exposing participants to unacceptable risks [23]. Beyond physical safety, WoZ is also methodologically justified in contexts where privacy or ethical concerns make the deployment of real data-collecting systems impractical [160].

Guideline 2: choose WoZ for experimental rigor and causal analysis.

For hypothesis-driven research aiming to isolate specific interaction variables, WoZ offers superior internal validity compared to stochastic automated systems. This choice reflects a strategic trade-off within McGrath's "Three-Horned Dilemma" [96], which posits that research strategies cannot simultaneously maximize precision, realism, and generalizability. For example, while deploying a real AI model might increase realism, it often introduces uncontrolled variance due to the probabilistic nature of algorithms. By employing a human wizard, researchers strategically prioritize precision to ensure rigorous causal analysis. Our findings confirm that researchers leverage WoZ to *Control Confounds*, such as strictly maintaining constant recognition rates [46] or injecting errors at

precise moments [176]. Consequently, when the epistemic goal is to establish causal links between design features and user experience, WoZ is methodologically preferable to using a live, unstable algorithm.

Guideline 3: leverage WoZ for interaction depth and benchmarking. In exploratory phases aiming to probe the upper bounds of user experience or elicit complex social behaviors, WoZ should be employed to simulate idealized capabilities. This approach is grounded in Suchman’s distinction between plans and situated actions [144]. While automated systems operate primarily on probabilistic “plans” derived from rules or training data, human wizards excel at “situated actions,” responding dynamically to the unique, ambiguous, and unfolding context of a specific interaction. Our findings show that researchers value this *Adaptive Flexibility* to handle socially nuanced inputs that rigid algorithms often fail to grasp. Li et al. [86] demonstrated that a human wizard acting as a doctor asked significantly more insightful follow-up questions than a GPT-4 powered agent, thereby uncovering deeper patient needs. By serving as a *Benchmark*, WoZ defines the human-level ceiling of interaction quality, helping researchers establish what the system should ideally do before engineering constraints dictate what it can do.

Our review also identified *Alignment with Prior Work* and *Cost-Efficiency* as common rationales for employing WoZ. However, we argue that neither constitutes a sufficient methodological justification on its own. First, relying on tradition ignores the shifting nature of technology. As we discuss in Section 5.1, the plausibility horizon and underlying technical grounding evolve continuously. A method appropriate in earlier technological contexts may no longer be appropriate today. Therefore, aligning with prior work is only valid if it operates alongside the methodological reasons discussed above. Used in isolation, it risks perpetuating outdated practices. Second, while recent advances in generative tooling and LLM-based prototyping substantially lower implementation barriers, relying solely on WoZ for efficiency is insufficient. As Fidelity-Cost trade-off theories [56] suggest, low-fidelity prototyping is only cost-effective if it yields valid data without incurring excessive overhead. Thus, using WoZ for convenience is methodologically untenable. The choice must be grounded in its specific capacity to yield data that automated alternatives cannot.

Finally, our review reveals a concern: 22.7% ($N = 44$) of WoZ papers did not specify any motivation for employing WoZ, which may also bias the apparent proportions of existing motivation categories. Omitting explicit motivations undermines transparency and hinders collective evaluation of the method’s contributions. We argue that future studies should explicitly articulate their rationale for employing WoZ, whether to overcome technical limitations, provide flexibility, ensure safety, or for other reasons.

5.2.2 Ensuring Rigorous and Responsible Use of WoZ. Once the decision to employ WoZ is made, the challenge shifts to method execution. Our review identified five categories of methodological concerns (Section 4.4) that can undermine data quality if left unmanaged. To support rigorous and responsible deployment of WoZ, we suggest that practitioners adhere to the following principles.

Principle 1: standardize and document wizard behavior.

One of the primary threats to internal validity is *Wizard Variability and Bias*, where inconsistent human performance introduces uncontrolled variance that can confound experimental manipulations. Furthermore, as Riek [122] noted, the frequent underreporting of wizard operations undermines *Transparency and Reproducibility*. While our review found that the majority of papers provided clear descriptions, a small subset (2.1%, $N = 4$) still described only system functions without explicitly specifying wizard actions. This calls for explicit decision protocols and rigorous training procedures aimed at minimizing behavioral variance and achieving consistent decision-making across wizards. Following Green et al. [44], researchers should specify behavioral hypotheses to constrain wizard actions. Crucially, publications should document and release the wizard’s operational log, including intervention criteria, decision rules, timing constraints, error rates, and training duration, with a level of granularity comparable to that expected for algorithmic descriptions.

Principle 2: optimize execution through load management.

While wizards offer flexibility, they face cognitive and physical limits, leading to *Temporal and Performance Constraints* such as noticeable latency or wizard fatigue. Riek also notes that “wizard production variables” (e.g., response speed) significantly impact believability [122]. To mitigate these bottlenecks, researchers should leverage load balancing. This includes employing a multi-wizard setup where different interaction tasks are distributed across different operators [179]. Furthermore, execution could be supported by tools. Beyond traditional pre-scripted response libraries [134], researchers can integrate generative models to draft content in real-time [15]. By offloading tasks to tools or collaborators, wizards can focus on the portions of the interaction they can manage consistently.

Principle 3: constrain actions and debrief ethically. The power of WoZ to simulate future intelligence carries the risk of breaking *Ecological Validity*, eliciting user behaviors or expectations that real systems could not reproduce. To ensure validity, simulations must be constraint-based. As emphasized by Fraser and Gilbert [32], valid simulations require a detailed specification of the target system’s behavior, including its limitations. Researchers should deliberately introduce artificial constraints, such as probabilistic errors [180] or bounded response strategies, to align the simulation with the plausibility horizon (Section 5.1) of the envisioned technology. Meanwhile, high-fidelity simulations increase the risk of participant *Deception*, which should be carefully managed through post-study debriefing [139].

5.3 WoZ in the LLM Era

The framework of reciprocal evolution (Section 5.1) positions WoZ not as a static method, but as one that adapts alongside technological shifts. Today, LLMs represent a predominant force in this landscape, expanding the plausibility horizon with unprecedented generative capabilities that intersect with tasks traditionally simulated by human wizards. We argue that this partial obsolescence does not imply the method’s demise. In this subsection, we examine LLMs as a contemporary case study to observe the reciprocal evolution framework in action, focusing on the complementary division



Figure 3: Mapping between WoZ application domains (our review, left) and LLM-ification [113] domains (right).

of labor between WoZ and LLMs, emerging strategies for integrating them in practice, and how rapid advancements are impacting the plausible space.

5.3.1 Complementary Niches of WoZ and LLMs. To examine the distinct applications of WoZ and LLMs, we mapped our identified domains against those reported in *LLM-ification of CHI* [113], a recent systematic review of 153 LLM-related papers at CHI. Their study also categorized applications into ten domains, ordered by frequency of occurrence. Figure 3 illustrates this mapping.

The first divergence emerges in the domain of *Work & Creativity Support*, which accounts for the largest share of WoZ studies yet has only a limited counterpart in the LLM-ification taxonomy, represented by the comparatively smaller categories of *Design* and *Creativity*. This imbalance suggests a methodological distinction. WoZ has long been valued for supporting early-stage ideation and collaborative making, where human wizards can flexibly interpret open-ended user input, mediate group dynamics, and sustain interactional flow in ways that immature algorithms cannot. As a result, WoZ has been central to prototyping co-creation scenarios [118], embodied design workshops [148], and collaborative performance systems [165]. In contrast, LLM-based research in this space remains narrower, often focusing on discrete tasks such as assisting individual designers with UI generation [27] or supporting text-based creativity [11].

A second major divergence lies in those embodied and situated interaction domains, including *HRI*, *Multimodal & Emerging Techniques*, *Mobile & Ubiquitous & Wearable Computing*, *Automotive User Interfaces*, and *XR/VR/AR*. Collectively, these account for more than half of all WoZ studies. In these settings, human wizards can flexibly orchestrate multimodal inputs, simulate complex sensorimotor capabilities, and preserve participant safety while approximating autonomous operation. By contrast, the *LLM-ification of CHI* review subsumes such work under the broader category of *Augmenting*

Capabilities, which represents only about 17% of LLM-related studies and primarily emphasizes productivity support or sensemaking tasks rather than embodied interactions [113].

Conversely, certain areas exhibit the reverse pattern: LLMs have quickly taken the lead in *Communication & Writing*, *Responsible Computing*, and *Programming*, domains that WoZ historically engaged with only marginally. These areas align closely with the intrinsic strengths of LLMs: natural language generation, large-scale text processing, and algorithmic reasoning. In *Communication & Writing*, for instance, LLMs directly instantiate the autonomous content generation that WoZ could only simulate, enabling empirical studies on AI-mediated communication [33, 47] and collaborative writing [162] at scale. Similarly, *Responsible Computing* and *Programming* foreground technical and ethical challenges specific to computational artifacts.

Beyond domains, WoZ and LLMs also differ in how they afford experimental control. LLMs are inherently probabilistic. Their outputs rely on opaque parameters that are difficult to isolate as independent variables. By contrast, WoZ affords fine-grained and inspectable control over system behavior. As suggested in 5.2.2, every rule, prompt, and decision can in principle be explicitly designed, constrained, and logged. This makes WoZ particularly suitable when researchers need precise manipulation of independent variables or traceable decision processes, even in tasks that LLMs could technically perform.

Viewed through the reciprocal evolution's perspective, these complementary niches also implicate directions for future LLM-based prototyping. We expect LLMs to expand into domains where WoZ is currently dominant. This expansion will happen as models gradually learn to handle the specific wizard control types we identified in Section 4.2. *Pre-scripted Content Control* and parts of *Communication Control* are already well aligned with current LLM strengths. Meanwhile, researchers are beginning to apply multimodal models for *Embodied and Environmental Action Control* [58, 98]. Furthermore, *System Logic Control* connects directly to the rise of autonomous agents, where the model functions as a reasoning engine to perceive contexts and plan decisions [170]. Finally, *Wizard Collaboration and Mediation* presents a more complex challenge, pointing toward future work in wizard-LLM collaboration (which we will discuss in Section 5.3.2), and even multi-agent orchestration [152].

5.3.2 Co-use Patterns of WoZ and LLMs. Having clarified the technological grounding provided by LLMs, we observe that currently WoZ and LLMs often operate in distinct niches. At the same time, the emerging co-use of wizards and models is actively revealing novel plausible space. Although studies combining WoZ and LLMs remain relatively limited ($N = 13$), our inductive analysis of this subset identifies several preliminary recurring patterns.

Wizard with LLM (53.8%, $N=7$). This pattern refers to systems in which WoZ and LLMs jointly contribute, with each responsible for a distinct part of the interaction pipeline. Across these systems, wizards occupy complementary positions around the model. There are wizards acting on the *input side*, organizing messy user intent into structured prompts that models can process effectively [15]. Wizards can also serve on the *output side*, curating or refining AI-generated suggestions before they reach the user [45]. More

separate collaboration includes wizards handling complex queries that exceed the model’s competence while delegating routine tasks to the AI [75].

In terms of reciprocal evolution, these systems illustrate a first reallocation of responsibilities once LLMs are introduced. Models take over much of the low-level generative work that wizards previously produced manually, while WoZ migrates toward curating, constraining, and selectively overriding model behavior. Our findings show this hybrid emerges most prominently in *Work & Creativity Support* domains. The prominence of this pattern arises from the nature of creative and complex problem-solving tasks, which calls for a combination of generative fluency and strategic oversight beyond what humans or LLMs can achieve individually.

However, most of these early instances of *Wizard with LLM* primarily function as a reliability bridge. Wizards repair missing context, screen hallucinations, and handle edge cases that the model cannot safely address, rather than using WoZ to enact qualitatively new capabilities. Even in systems that imagine more context-aware agents [15], the wizard’s work largely orients around organizing LLM inputs. Future research could increasingly use this co-use pattern not just to patch limitations, but to explore broader plausible space. For instance, wizards could simulate agents with perfect long-term memory, proactive initiative, or deep personalization. Such uses would more fully realise the reciprocal evolution framework by using WoZ to project interactions into futures that current LLMs cannot yet support, in turn informing where technical development should next extend the plausibility horizon.

Wizard towards LLM (23.1%, N=3). This pattern refers to using WoZ as a preliminary design probe that informs and guides the development of subsequent LLM-based systems. Rather than substituting for unavailable technical functions, the wizard here enables researchers to elicit authentic practices and derive theoretical or design frameworks that later guide the construction of LLM-driven prototypes. Researchers have leveraged this approach to extract persuasion strategies for digital well-being [169], prioritize interface signals for hallucination detection [83], and uncover implicit user expectations for generative VR scripting [1]. In these workflows, the insights derived from wizard-mediated interactions, such as specific mental states or preferred prompting styles, serve as the ground truth that shapes the prompts, architectures, and fine-tuning targets of the subsequent LLM implementations.

Our review may under-report this pattern, since two-phase studies are often split across separate publications. Even so, this pattern operationalizes the cycle of reciprocal evolution by ensuring that desirability leads feasibility. Instead of letting model capabilities dictate interaction design, the wizard actively charts the plausible space, generating the behavioral ground truth that steers the subsequent expansion of the plausibility horizon.

Wizard versus LLM (23.1%, N=3). This pattern uses the wizard as a benchmark to evaluate LLM performance. The wizard acts as the upper bound of interaction quality, representing expert human judgment. Recent studies use this comparison to test if models have truly caught up to a human level. For example, in proactive AR tasks, LLMs have achieved performance comparable to human experts [87]. However, in complex settings like medical interviews, human wizards still outperform models, asking significantly more

follow-up questions and gathering richer patient information than chatbots [86].

This pattern serves as a critical calibration mechanism. While early WoZ studies compared wizards against rudimentary interfaces [46, 65, 67] to prove utility, today’s comparisons serve a stricter purpose: to verify if the technology has truly reached the plausibility horizon. If so, the frontier can be further pushed outwards. However, wizard-based comparisons also present methodological concerns. Wizard performance varies by operator and context, making it difficult to treat as a fixed upper bound. In some cases, the model may outperform a given wizard in consistency or efficiency [29]. This calls for treating wizarding as a standardized protocol (Section 5.2), explicitly documenting wizard training, decision rules, and consistency checks.

These insights are also relevant for communities that design evaluations for AI systems. WoZ-baselines help evaluate model performance in ways that static datasets cannot. During evaluation, researchers need to state how human performance is defined, enacted, and constrained. Such reflexive reporting can guide the design of evaluation protocols in AI and related fields, where model behavior is interpreted against transparent human reference points.

5.3.3 Reframing the Plausible Space. The rapid advancement of LLMs introduces new challenges. From the perspective of our framework in Section 5.1, LLMs have pushed the inner boundary of what can be implemented so fast. At the same time, it has become difficult for both users and researchers to estimate what these models can and cannot reliably do. As a result, the plausibility horizon is unstable, and the plausible space between implementation and belief becomes fleeting. Features that require simulation today may be immediately automated tomorrow. Furthermore, as models approach or exceed human performance in specific tasks (e.g., knowledge retrieval or coding), the wizard can no longer credibly simulate a "better" performer.

This highlights a broader design space in which wizarding helps to re-establish and probe the plausible space in the LLM era. For example, use wizards and LLMs together to map the moving boundary of plausibility. Wizards can deliberately stage interactions that assume stronger memory, deeper personalization, or richer context knowledge than current models can sustain, then observe where users still find the system credible and where they begin to doubt it. These observations can clarify how far the plausibility horizon extends beyond current LLM capabilities and which gaps matter most for experience.

Finally, the dilemma also connects to the boundary conditions of our framework in Section 5.1. There we noted that as technologies advance toward more general or even super-human performance, the reciprocal evolution framework may no longer hold. In these settings, the challenge shifts from simulating missing system capabilities to elevating human capabilities, considering how humans can harness such intelligence while maintaining agency and ethical control. An extended plausible space that includes not only new system behaviors but also new forms of human augmentation could be considered. Wizards and LLMs can jointly prototype how control, responsibility, and decision making are distributed, and how different configurations affect user agency. Exploring how to

support a stable plausible space when technical capabilities change rapidly is an open agenda for future work.

5.4 Limitations & Future Work

As with any systematic review, our study has several limitations that should be acknowledged to contextualize the scope and validity of our findings. First, our search strategy, limited to WoZ-related keywords, may have excluded studies that described similar practices under different terms (e.g., "Ghost Driver" in autonomous vehicle research [125]). Through snowballing, we also found that some papers employing the WoZ method did not explicitly mention it in their titles, abstracts, or keywords, further indicating potential gaps in coverage. Moreover, as our corpus focuses on SIGCHI publications, the findings may not fully capture WoZ practices in adjacent fields. Second, the observed growth in WoZ publications may partly reflect the expansion of SIGCHI venues and the increasing number of accepted papers, rather than a genuine surge of methodological interest. Third, although we employed multi-coder procedures with IRR reported, some coding decisions (e.g., for control types or motivations) remained open to interpretation and thus involved researcher judgment. Our coding scheme, while systematically developed, may not have exhausted all possible dimensions of WoZ use, and alternative categorizations could reveal additional insights. We also note a potential techno-centric bias in our reciprocal-evolution framing. Emphasizing technology–method co-evolution risks reducing "plausibility" to technical reachability, while under-specifying the socio-systemic conditions that would make a future interaction viable (or contestable) in practice [129]. In addition, the discussion of WoZ–LLM co-use was based on a small and temporally clustered sample ($N = 13$), limiting the generalizability of observed patterns. As part of our future work, we plan to update the corpus as more studies emerge, thereby enabling more systematic validation of these categories and patterns.

6 Conclusion

This paper provides the first systematic literature review of WoZ practices across HCI, drawing on 194 papers published in SIGCHI venues. By synthesizing evidence from over three decades of research, we identified ten application domains, five types of wizard control, eight distinct motivations for employing WoZ, and five categories of concerns along with corresponding mitigation strategies. Going beyond these classifications, our analysis reveals a reciprocal evolution between technology and wizarding. As technical capabilities expand the horizon of what is possible, WoZ shifts to explore the plausible interaction space beyond current feasibility. To support this evolving role, we propose practical guidelines that match WoZ to specific research goals, emphasizing rigorous execution and transparency. Furthermore, we instantiate this framework through the lens of LLMs, discussing the divergence of applications and the emerging patterns of WoZ–LLM co-use. Looking forward, the insights from this review can inform the design of tools and reporting standards for wizarding, guide the study of new configurations that mix human and automated components, and support the responsible integration of WoZ into emerging AI- and LLM-based prototyping workflows.

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